

Statistical Assessment of Ionospheric Perturbation Amplitudes Caused by Tohoku-Oki Earthquake in Japan

Kiyoung Sun, Moonseok Yoon, and Jiyun Lee
Department of Aerospace Engineering, KAIST, Daejeon, Republic of Korea
sunstheory@kaist.ac.kr

Introduction

Natural hazards such as earthquakes, tsunamis and volcanic eruptions have been significant threats, causing tremendous casualties and property damages. Natural hazards release a huge amount of energy for only a short period of time, and they generate acoustic-gravity and sound waves into the atmosphere [1]. These waves reach to ionosphere and result in Traveling Ionospheric Disturbances (TIDs). The properties of TIDs such as amplitude, period, velocity vary depending on the form or scale of natural hazards [1][2][3].

The ionospheric remote sensing technique using Global Navigation Satellite System (GNSS) data provides world-wide observation coverage with widely distributed their networks. While taking this advantage, many researchers examine the feasibility of GNSS-based natural hazard monitoring/detecting system [2][3]. As a part of these works, this paper assesses amplitude of ionospheric perturbations caused by the *M* 9.0 Tohoku-Oki earthquake event on the March 11th, 2011 using GNSS data. In addition, we examine the potential of natural hazard monitoring using TID perturbation amplitudes by comparing the resulting statistics to those under nominal ionospheric conditions.

Data

Potentially nominal ionospheric days were selected based on geomagnetic indices, Planetary Kp (*Kp*) and disturbance storm time (*Dst*). The Kp index represents the solar particle effects on the earth and it is a composite index measured over a 3 hours period at several mid-latitude stations primarily located in the northern hemisphere [4]. The Dst index measures equatorial magnetic disturbance derived from hourly scaling of low-latitude horizontal magnetic variation [5]. As Kp and Dst in Table 1 indicate, the geomagnetic conditions in 7 consecutive days were relatively quiet.

Table 1 Days selected to conduct stochastic analysis of ionospheric perturbations under nominal ionospheric conditions

Date (UT yy/mm/dd)	Kp	Dst
15/01/14	2.7	-24
15/01/15	3.0	-22
15/01/16	3.0	-14
15/01/17	3.0	-14
15/01/18	2.3	-21
15/01/19	2.3	-11
15/01/20	1.3	-15

GNSS signal data ($L_1 = 1575.42$ MHz and $L_2 = 1227.60$ MHz) on selected nominal days and March 11th, 2011 were obtained from total 66 GPS Earth Observation Network (GEONET) stations located within 250 km from the epicenter (38.322 °N, 142.369 °E). Obtained GNSS observation data were input into Long-Term Ionospheric Anomaly Monitor (LTIAM) (see [6]) to estimate Total Electron Content (TEC) values which corresponds to L1 signal.

Methodology

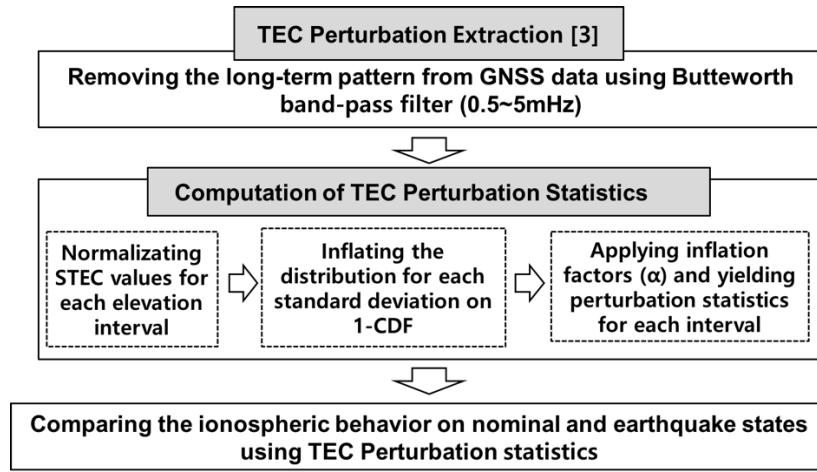


Figure 1. Flow chart of assessing ionospheric perturbation statistics

A. Ionospheric Perturbation Estimation

The whole process of this analysis is described in Figure 1. The ionosphere has a long-term pattern of diurnal/seasonal trend and it gradually changes over a wide area. If natural hazards such as earthquakes, tsunamis and volcanic eruptions occur, the perturbations are generated on the ionosphere in the form of TIDs. Thus, TID signatures can be extracted with removing the long-term pattern from GNSS data.

In this study, we used a Butterworth band-pass filter for TIDs extraction with a passband of 0.5 ~ 5 mHz [3]. A passband of the filter was determined with wave-front periods of tsunami measurements [3]. Using this technique, ionospheric perturbations were extracted from the TEC time series for each station-satellite pair.

B. Computation of ionospheric perturbation statistics

After extracting the ionospheric perturbation for every satellite-station pair, we normalized residual TEC values (or amplitude) with means and standard deviations of divided elevation bins. Note that satellite elevations angles were divided as 5 bins; 30° ~ 40°, 40° ~ 50°, 50° ~ 60°, 60° ~ 70° and 70° ~ 90°. Then, we obtained a one-minus Cumulative Distribution Function (1-CDF). Since actual perturbation distribution has non-Gaussian tails violating the 1-CDF of standard normal distribution. Thus, the one-sigma of normalized perturbations was dilated by multiplying inflation factor (α) to cover the non-Gaussian tails of actual distribution. This paper determined the inflation factor as 1.7020. As shown in Figure 2, the resulting 1-CDF distribution after applying the inflation factor is well bounded by the standard distribution. The same process is conducted for the nominal ionospheric days. The ionospheric perturbations statistics $\sigma_{\text{filtered STEC}}$

overbounds, were computed as $\left| \mu_{\text{filtered STEC}} \right| + \alpha \sigma_{\text{filtered STEC}}$.

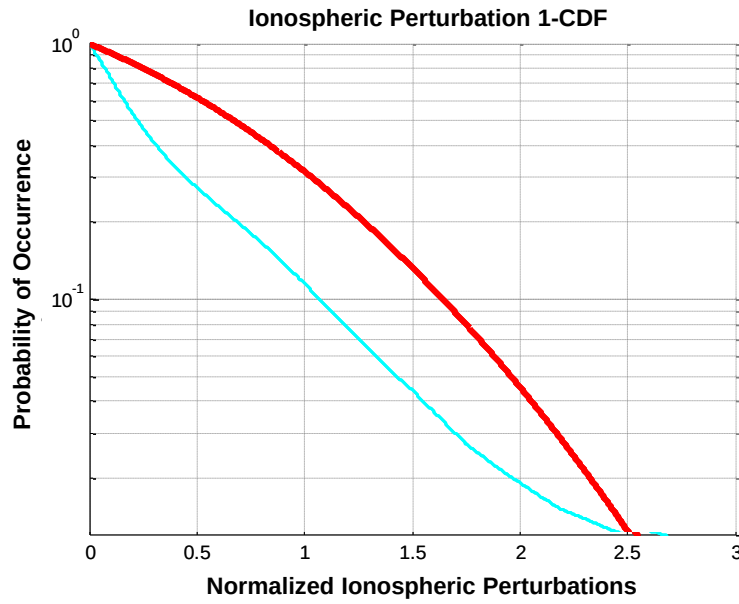


Figure 2. Inflation factor-applied 1-CDF distributions of ionospheric perturbations for two hours right after the earthquake (blue sky solid line) and 1-CDF of standard normal distribution (red solid line)

Results

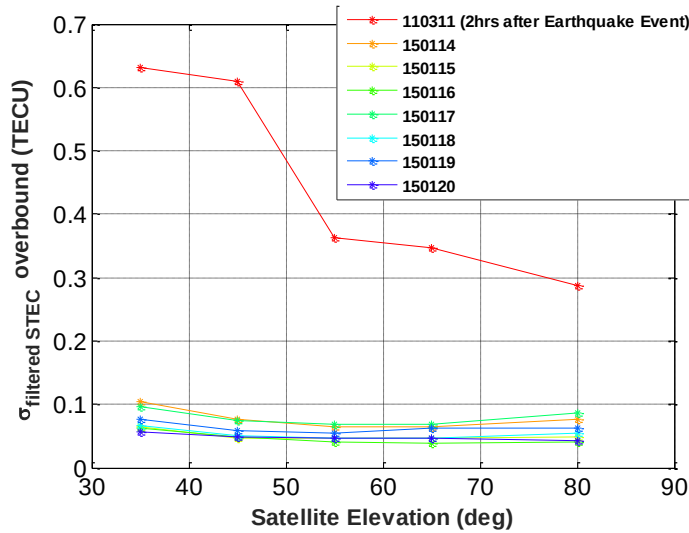


Figure 3. Ionospheric perturbation statistics $\sigma_{\text{filtered STEC}}$ for each elevation bin.

As you can see in Figure 3, $\sigma_{\text{filtered STEC}}$ overbounds of seven days in nominal ionospheric conditions have approximately 0.04 ~ 0.1 TECU over entire elevation bins. However, $\sigma_{\text{filtered STEC}}$ overbounds during 2 hours right after the earthquake have 0.29 ~ 0.63 TECU. Thus, the ionospheric behavior caused by the earthquake is totally different from those in nominal ionospheric conditions. Therefore, based on the results of this case study, the natural hazard monitoring using TID perturbation amplitudes can be considered to be practicable.

Conclusions

In this study, we examine the ionospheric behaviors caused by the *M* 9.0 Tohoku-Oki earthquake event in Japan by

comparing the TID statistics to those under nominal ionospheric conditions. We confirmed that the results from two different data set showed significant difference. Based on the results of the monitoring feasibility, the monitoring and detecting algorithms for anomalous TID amplitudes caused by natural hazards should be designed through further studies.

Acknowledgement

Kiyoung Sun was supported by Undergraduate Research Participation Program funded by the Korean Advanced Institute of Science and Technology (N01150503).

References

- [1] Galvan D.A., A. Komjathy, and M. P. Hickey, *Ionospheric signatures of Tohoku-Oki tsunami of March 11, 2011: Model comparisons near the epicenter*, Radio Sci., 47, Rs005023", 2012
- [2] Yang, Y., Komjathy A., Butala M., Mannucci A. J., Snively J., Hickey M., Galvan D., Lee J., and Langley R. B., *Ionospheric Natural Hazard Detection Using GNSS Measurements: Investing the Chelyabinsk Meteor Impact*, ION GNSS 2013, 2013
- [3] Komjathy, A., D.A. Galvan, P. Stephens, *Detecting Ionospheric TEC Perturbations caused by natural hazards using a global network of GPS Receivers: The Tokyo Case Study*, 2012
- [4] Menvielle M., and A. Berthelier, *The K-derived planetary indices: Description and availability*, Rev. Geophys., 29(3), 415-432, doi:10.1029/ 91RG00994, 1991
- [5] Sugiura M., and T. Kamei, *Equatorial Dst index 1957-1986*, IAGA Bull., 40, 7-14., 1991
- [6] Jung S., and J. Lee, *Long-term ionospheric anomaly monitoring for ground based augmentation systems*, Radio Sci., 47, RS4006, doi:10.1029/2012RS005016, 2012